

ISAAC (ZAC) ADJEI

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[Date]

[Hiring Manager Name/Recruitment Team]

[Company Name]

[Company Address/Location]

Re: Application for [Role Title] at [Company Name]

Dear [Hiring Manager Name/Recruitment Team] ,

I am writing to apply for the [Role Title] position at [Company Name] . As an Electronic Engineering and Computer Science student at Aston University (Predicted First Class) and Top 40 Finalist in the Black Heritage Undergraduate of the Year Award 2026, I have built end-to-end data pipelines, applied unsupervised machine learning to real sensor data and written production ETL scripts that run in CI: all outside of coursework, on systems I operate myself.

The centrepiece of my data work is the PHAEMOS platform, where I designed and implemented an Isolation Forest anomaly detection model trained on 18-stream telemetry from a 4-node IoT platform (ESP32, STM32, Arduino Nano, Raspberry Pi Pico 2W). The pipeline ingests raw ADC and vibration samples, engineers rolling features (mean, variance, rate of change over configurable windows), fits the model and flags anomalous readings with confidence scores: all within a FastAPI backend that maintains sub-200ms response times under load. I built every layer: the schema, the feature engineering logic, the model training workflow and the alerting integration. I also completed a virtual engineering operations placement with British Airways, producing a data-driven Material Forecast and Planning Report for six A320 aircraft, including long-lead part risk analysis and scenario modelling under EASA/CAA compliance constraints. I am drawn to [Company Name] because of [specific reason] and I am keen to work on [specific data challenge or product] .

My data engineering instincts come from building systems where bad data has visible consequences: a miscalibrated sensor trips a false alert. That real-world pressure trained me to validate at ingestion, log failures explicitly and design pipelines that degrade gracefully rather than fail silently. I am fluent in Python (pandas, numpy, scikit-learn, requests), comfortable in SQL (Supabase PostgreSQL, complex window functions, upsert patterns) and experienced with Supabase PostgREST for lightweight data API exposure.

Data problems interest me most when the data is messy: inconsistently formatted, partially missing or arriving from sources with no documented schema. [Company Name] 's work on [specific data product or pipeline challenge] is the kind of problem I find genuinely engaging, and I would bring both the engineering rigour and the curiosity to tackle it systematically.

Thank you for considering my application. I would welcome a conversation about how my data engineering and ML experience can contribute to [Company Name] .

Yours sincerely,

Isaac (Zac) Adjei